



MANIPAL ACADEMY OF HIGHER EDUCATION

(Deemed-to-be-University under Section 3 of the UGC Act, 1956)

Information Security Lab

V Semester: B. Tech IT/CCE (ICT 3141)

Year: 2025

SCHOOL OF COMPUTER ENGINEERING

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

MANIPAL INSTITUTE OF TECHNOLOGY

Manipal – 576 104

SCHOOL OF COMPUTER ENGINEERING



Certificate

This is to certify that Ms./Mr.

Reg. No. Section: Roll No: has
satisfactorily completed the lab exercises prescribed for Information Security Lab [ICT
3141] of Third Year B. Tech. (IT/CCE) Degree at MIT, Manipal, in the academic year
2025-2026.

Date:

Signature of the faculty

CONTENTS

Lab No.	Title	Page No.	Marks	Remarks	Sign
	Course Objectives, Outcomes and Evaluation Plan				
	Instructions to the Students				
1	Basic Symmetric Key Ciphers	1			
2	Advanced Symmetric Key Ciphers	6			
3	Asymmetric Key Ciphers	14			
4	Advanced Asymmetric Key Cryptography	22			
5	Hashing	31			
6	Digital Signature	34			
7	Partial Homomorphic Encryption	40			
8	Searchable Encryption	48			
9	Project Design	56			
10	Project implementation	57			
11	Testing and validation	58			
12	References	59			

LAB NO.: 10

Date:

PROJECT IMPLEMENTATION

Objectives:

- To analyze the working of the front end and back end altogether.

Lab exercises:

1. Give the implementation details regarding the project.

[OBSERVATION SPACE – LAB 10]

LAB NO.: 11

Date:

TESTING AND VALIDATION

Objective:

- To test and validate their mini project.

Lab Exercises:

1. Give the testing and validation details for your project.

[OBSERVATION SPACE – LAB 11]